Alternating Current

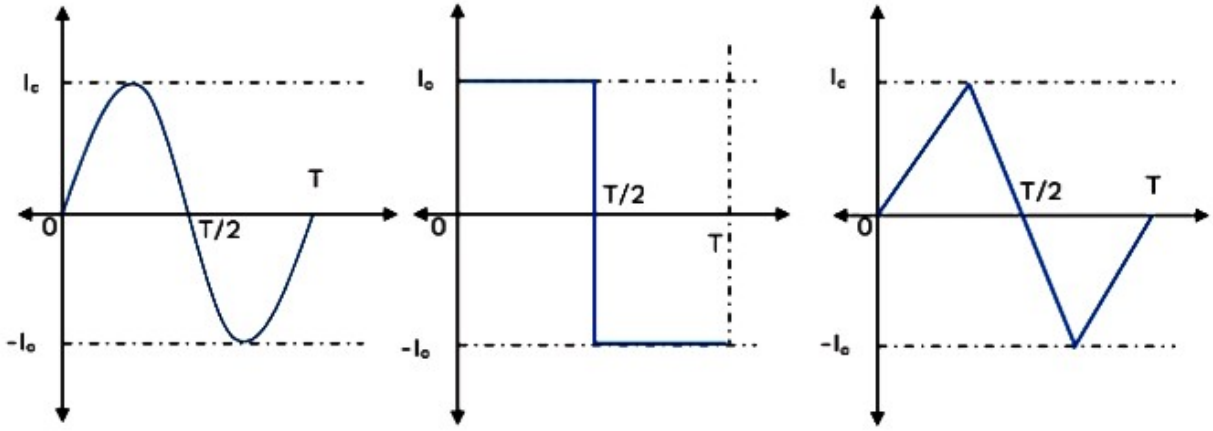
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Alternating Current

Alternating Current: A current whose direction of flow changes periodically is known as alternating current. For half cycle of the period, the current flows in one direction and for the other half cycle, it flows in opposite direction.

The following graphs represent some alternating currents:



Sinusoidal A.C. Square wave A.C. Triangular wave A.C.

Sinusoidal AC emf and its Generation:

When a coil is rotated at uniform angular speed 'ω' in a magnetic field, an emf is induced in the coil.

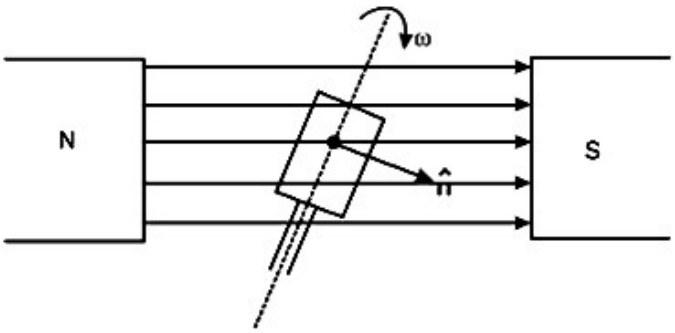
Magnetic flux linked to coil = NBA cos ωt

$$e = \left|\frac{d\phi}{dt}\right| = NBA\omega \sin\omega t$$

$\omega t = \theta$ is the angle between $\vec{B}$ and $\hat{n}$.

Here $NBA\omega$ =maximum emf = E_o

So, induced emf, $e = E_o \sin \omega t$:



Average value of an AC:

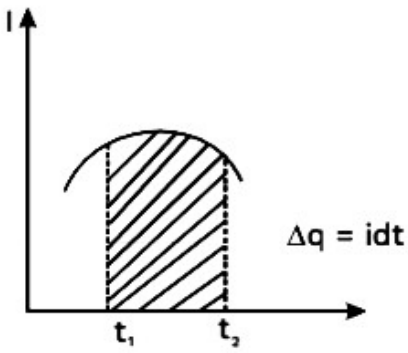
Current represents the rate of flow of charge.

Average value of a.c. is the total charge that flows through a circuit element in a given time interval divided by that time interval.

Average value of current in circuit $I_{avg} = \frac{\Delta q}{\Delta t}$,

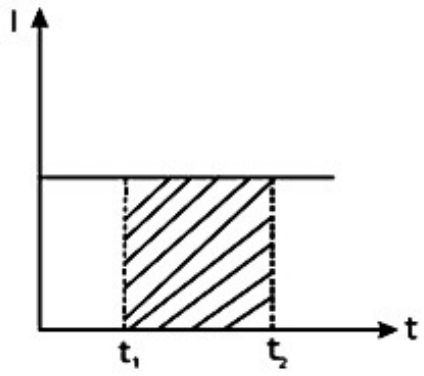
Where Δq represents the charge flown in time interval Δt.

Average current for a time varying current is given by,



$$I_{avg} = \frac{\int_{t_1}^{t_2} I\,dt}{(t_2 - t_1)} = \frac{1}{\Delta t}\int_{t_1}^{t_2} I\,dt$$

In case of D.C. [Direct current]



$I_{avg} = I_{dc}$, because this current will not change with time.

As $I_{avg} = \frac{1}{\Delta t}\int_{t_1}^{t_2} I\,dt$

Mean or average value of AC is equal to that DC value due to which in a circuit same charge flows, which flows due to AC in same duration.

Charge flow due to DC = $I_{dc}[t_2 - t_1]$

Charge flow due to AC = $\int_{t_1}^{t_2} I dt$

If both are equal as per definition, then

$$I_{dc}(t_2 - t_1) = \int_{t_1}^{t_2} I\,dt$$

As $I_{dc} = I_{avg}$

So, $I_{dc} = I_{avg} = \frac{\int_{t_1}^{t_2} I\,dt}{t_2 - t_1}$

Root mean square value of an AC:

For a time varying function, average value $< I(t) >= \frac{\int_{t_1}^{t_2} i\,dt}{t_2 - t_1}$

Now, If we calculate the root mean square value of time varying function, we can write,

$$I_{rms} = \sqrt{\frac{\int_{t_1}^{t_2} i^2\,dt}{\int_{t_1}^{t_2} dt}}$$

Significance of RMS current:

For a time varying current i_{ac} (t), the total heat produced when it is passed through a resistance 'R' in time t_1 to t_2 is:

$$H_1 = \int_{t_1}^{t_2} I_{ac}^2 R dt$$

If dc current is passed through the same resistance for same time interval, the heat produced is, $H_2 = I_{dc}^2 R(t_2 - t_1)$

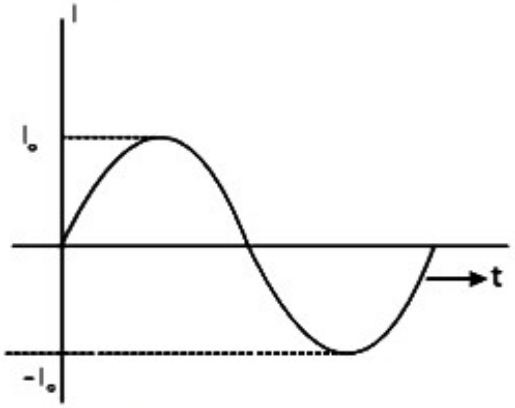
If both $H_1 = H_2$, then

$$I_{rms} = \sqrt{\frac{\int_{t_1}^{t_2} i_{ac}^2\,dt}{t_2 - t_1}}$$

RMS value of given AC can be defined as that DC value which produces same heat in a resistance in the same interval of time as produced by AC from the same resistance in same interval of time.

Average current of a Sinusoidal A.C. :

As the sinusoidal form of A.C is, $I = I_0 \sin \omega t$



For one complete cycle, mean value of current is zero as there is a reversal in the direction of current after every half cycle.

$\therefore I_{avg} = 0$

Now we calculate I_{avg} for half cycle.

$$I_{avg}\text{ (for half cycle)} = \frac{\int_0^{T/2} i\,dt}{\frac{T}{2}}$$

$$= \frac{2}{T}\int_0^{\frac{T}{2}} I_0 \sin\omega t$$

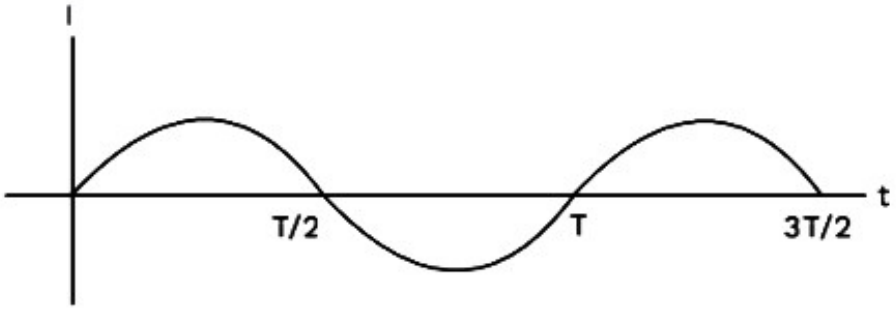
$$= \frac{2}{T\omega} I_0 \left[-\cos\omega t\right]_0^{\frac{T}{2}}$$

$$= \frac{2I_0}{2\pi}\left[-\cos\frac{\omega T}{2} + \cos 0\right] \quad .....\left(\text{as } \omega = \frac{2\pi}{T}\right)$$

$$= \frac{I_0}{\pi}\left[-\cos\pi + \cos 0\right] = \frac{I_0}{\pi}(2)$$

$$= \frac{2}{\pi} I_0 = 0.63\ I_0 \quad \text{or} \quad 63\%\ I_0 = (I_{avg})_{\text{Half cycle}}$$

RMS value of Sinusoidal AC current :



Sinusoidal current, $I = I_0 \sin\omega t$

As we know, $I_{rms} = \sqrt{\frac{\int_0^T i^2 dt}{T}}$

So, $$i^2_{rms} = \frac{1}{T}\int_0^T i_o^2 \sin^2\omega t\,dt$$

$$= \frac{i_0^2}{2T}\int_0^T (1 - \cos 2\omega t)\,dt$$

$$i^2_{rms} = \frac{i_0^2}{2T}\int_0^T \left[\left(T - \frac{(\sin 2\omega t - \sin 0)}{2\omega}\right)\right]$$

As we know that $\omega t = 2\pi$ and $\sin 4\pi = 0$,

$$i^2_{rms} = \frac{i_0^2}{2T}\left[T - 0\right] = \frac{i_0^{\,2}}{2}$$

Then, $i_{rms} = \frac{i_0}{\sqrt{2}} = 0.707\, i_0 = 70.7\%\ i_0$

Q. **An electric bulb is designed to operate at 12V DC. If the bulb is connected to an AC source and gives same brightness, what would be the peak voltage of AC.**

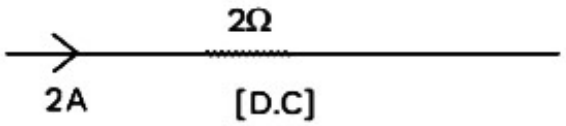
Sol. As brightness is same, that means effective (RMS) voltage of AC must be 12V.

$$e_{rms} = \frac{e_0}{\sqrt{2}}$$

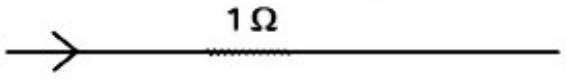
Peak value $(e_0) = \sqrt{2}\, e_{rms} = 12\sqrt{2}$volt

Q. **A direct current of 2A and an AC of peak value 2A flows through resistances of 2Ω and 1Ω respectively. Find the ratio of heat produced in the two resistances in same time interval.**

Sol.



$$H_1 = I^2Rt = 2^2\,(2)\,t = 8t$$



$$I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}A$$

$$H_2 = i^2_{rms}\, Rt = (\sqrt{2})^2 \times 1 \times t = 2t$$

So, $\frac{H_1}{H_2} = \frac{8t}{2t} = 4$

Q. **In a wire, a direct current I_1 and an AC current $I_2 \sin\omega t$ is superposed. Find the RMS current in wire.**

Sol. Total current in the wire

$$I = I_1 + I_2 \sin\omega t$$

As we know, $I_{rms} = \sqrt{\frac{\int_{t_1}^{t_2} I^2 dt}{\int_{t_1}^{t_2} dt}}$

First, we calculate the value of $\int_{t_1}^{t_2} I^2 dt$ ..... (i)

As we know the current is,

$$I = I_1 + I_2 \sin\omega t$$

$$I^2 = \left(I_1 + I_2 \sin\omega t\right)^2$$

$$= I_1^2 + I_2^2 \sin^2 \omega t + 2 I_1 I_2 \sin\omega t \quad \text{..... (ii)}$$

Now put the value of (ii) in (i)

$$\int_0^T I^2 dt = \int_0^T \left(I_1^2 + I_2^2 \sin^2 \omega t + 2 I_1 I_2 \sin\omega t\right) dt$$

$$= \int_0^T I_1^2 dt + I_2^2 \int_0^T \sin^2 \omega t \, dt + 2 I_1 I_2 \int_0^T \sin\omega t \, dt$$

As we know, $\int_0^T \sin\omega t \, dt = 0$ and $\int_0^T \sin^2 \omega t \, dt = \frac{T}{2}$

$$\therefore \quad \int_0^T I^2 dt = I_1^2 T + I_2^2 \left(\frac{T}{2}\right) \quad \text{..... (iii)}$$

Now put (iii) in the expression of i_{rms}

$$I_{rms} = \sqrt{\frac{I_1^2 T + I_2^2 \frac{T}{2}}{T}}$$

$$= \sqrt{I_1^2 + \frac{I_2^2}{2}}$$

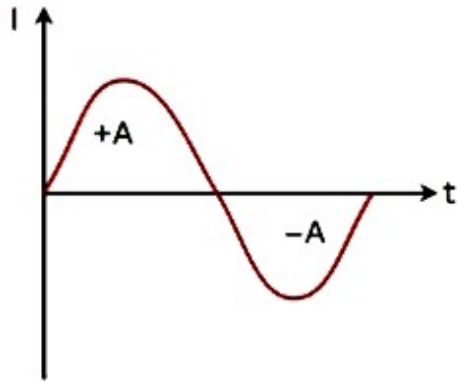
Hot Wire Instruments

The electrical instruments working on the principle of magnetic effects of current (like moving coil galvanometer) can only measure the mean or average value of an a.c. which is zero. So, these instruments are not suitable for measuring alternating currents.

In order to measure the alternating currents or voltages, we use devices which work on principle of heating effects of current and read the rms values of the alternating current or voltage.

Q. **The rms value of current in a 50 Hz AC circuit is 5 A. The average value of AC current over a cycle is?**

Sol.



Average value of the current over 1 cycle = +A −A = 0 A

Q. **The instantaneous voltage for an alternating voltage with a frequency 50 Hz and peak voltage 300 V is**

Sol. Given that, frequency, f=50 Hz and peak voltage, $V_0 = 300\,V$

Instantaneous voltage, $V = V_0 \sin\omega t = V_0 \sin 2\pi ft = 300 \sin 100\pi t$

Q. **The equation of a.c. in a circuit is given by I = 50 sin 314t. What are the peak value and frequency of current?**

Sol. Here, $I = 50 \sin 314t$

Comparing with standard equation,

Peak value of current, $I_0 = 50$ A

And,

$$\omega t = 314t \Rightarrow 2\pi f = 314 \Rightarrow f = \frac{314}{2\pi} = 50 \text{ Hz}$$

Q. **Find the rms value of AC which when passed through a resistor produces heat, which is twice that produced by a constant current of $\sqrt{2}$ A in the same resistor.**

Sol. For rms value of AC = I_{rms}, the heat produced in resistor R in time t is

$$H_{AC} = I_{rms}^2 Rt \quad \ldots(i)$$

Heat produced by the constant current I in the same resistor R in the same time t is

$$H = I^2 Rt \quad \ldots(ii)$$

Dividing (i) by (ii),

$$\frac{H_{AC}}{H} = \frac{I_{rms}^2}{I^2} \quad \text{or} \quad I_{rms}^2 = I^2 \frac{H_{AC}}{H}$$

Given that, $H_{AC} = 2H$, $I = \sqrt{2}$ A

$$\therefore \quad I_{rms}^2 = \left(\sqrt{2}\,A\right)^2 \left(\frac{2H}{H}\right) = (2A)^2 \quad \text{or} \quad I_{rms} = 2A$$

Q. **If $V = 100\sin(100t)$ volt and $I = 100\sin\left(100t + \frac{\pi}{3}\right)$ mA are the instantaneous values of voltage and current, then the rms value of voltage and current are**

Sol. The instantaneous value of voltage,

$$V = 100\sin(100t)V$$

On comparing with standard equation, $V = V_0\sin(\omega t)$

we get, $V_0 = 100V$, $\omega = 100\,rad\,s^{-1}$

Rms voltage, $V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{100}{\sqrt{2}}V = 70.7V$

The instantaneous value of current,

$$I = 100\sin\left(100t + \frac{\pi}{3}\right)mA$$

On comparing with standard equation, $I = I_0\sin(\omega t + \phi)$

We get, $I_0 = 100\,mA$, $\omega = 100\ rad\,s^{-1}$

Rms current, $I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{100}{\sqrt{2}}mA = 70.7mA$

Q. **Find the rms current when an alternating current is given by $I = i_1\cos\omega t + i_2\sin\omega t$.**

Sol. $i = i_1\cos\omega t + i_2\sin\omega t$

$$i^2 = i_1^2\cos^2\omega t + i_2^2\sin^2\omega t + 2i_1i_2\sin\omega t\cos\omega t$$

$$= i_1^2\left[\frac{\cos 2\omega t + 1}{2}\right] + i_2^2\left[\frac{1 - \cos 2\omega t}{2}\right] + i_1i_2\sin 2\omega t$$

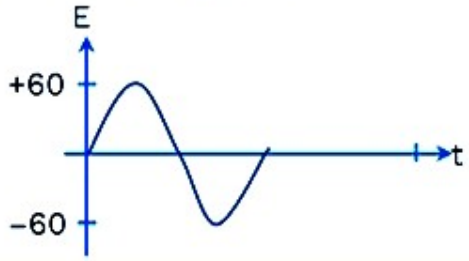
$$= \left(\frac{i_1^2 + i_2^2}{2}\right) + \left(\frac{i_1^2 - i_2^2}{2}\right)\cos 2\omega t + i_1 i_2 \sin 2\omega t$$

$$i_{rms} = \sqrt{\frac{\int i^2 dt}{\int dt}} = \sqrt{\frac{\left(\frac{i_1^2 - i_2^2}{2}\right)\int_0^T dt + \left(\frac{i_1^2 - i_2^2}{2}\right)\int_2^T \cos 2\omega t.dt + i_1 i_2 \int_0^T \sin 2\omega t.dt}{T}}$$

$$\sqrt{\frac{\left(\frac{i_1^2 - i_2^2}{2}\right)T + 0 + 0}{T}}$$

$$= \sqrt{\frac{i_1^2 + i_2^2}{2}}$$

Q. **The emf equation in an AC circuit is given by $E = 60\sin(100\pi t)$. Find the minimum time taken by emf to vary from +30 V to -30 V.**



Sol. $E = 60\sin(100\pi t)$ $\quad 30 = 60\sin(100\pi t_1) \Rightarrow \frac{1}{2} = \sin(100\pi t_1)$

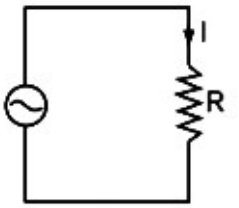
$$100\pi t_1 = \frac{\pi}{6} \Rightarrow t_1 = \frac{1}{600}\text{s}$$

$$-30 = 60\sin(100\pi t_2) \Rightarrow 100\pi t_2 = \frac{7\pi}{6} \quad \therefore \quad t_2 = \frac{7}{600}\text{s}$$

Minimum time taken, $t_2 - t_1 = \frac{7}{600} - \frac{1}{600} = \frac{6}{600} = \frac{1}{100}\text{s}$

A.C. VOLTAGE CONNECTED TO A RESISTOR

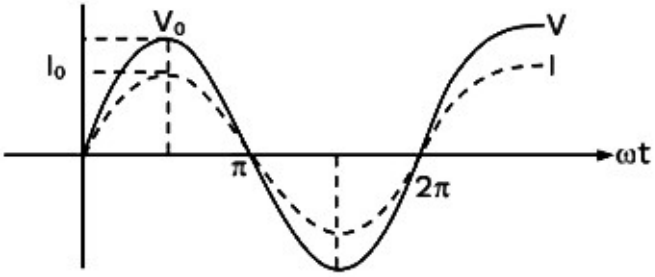
Let the A.C. voltage applied to the resistor be $V = V_0 \sin\omega t$ ...(i)
where V_0 is the maximum value of applied voltage and ω is the angular frequency.
By Kirchhoff's voltage law, $V_0 \sin\omega t = IR$.



$$\Rightarrow \quad I = \left(\frac{V_0}{R}\right)\sin\omega t = I_0 \sin\omega t \qquad \text{...(ii)}$$

where $I_0 = \left(\frac{V_0}{R}\right)$ is the amplitude of current.

From equation (i) & (ii), we can see that current and voltage are having the same phase, i.e. phase difference between them is zero.



For one complete cycle, average current will be zero.
Instantaneous Power in resistor, $P = I^2R = I_0^2 R \sin^2\omega t$

Average power in one cycle,

$$\bar{P} = \langle I^2R \rangle = \langle I_0^2 R \sin^2\omega t \rangle$$

$$\text{Also, } \langle \sin^2\omega t \rangle = \frac{\int_0^T \sin^2\omega t\, dt}{\int_0^T dt}$$

$$= \frac{1}{2T}\int_0^{T=\frac{2\pi}{\omega}} (1-\cos 2\omega t)dt = \frac{1}{2}$$

$$\text{So, } \bar{P} = \frac{I_0^2 \times R}{2}$$

The effective value of a.c. current for heat dissipation is equal to d.c. current which produces same heat as due to a.c. Thus,

$$\bar{P} = I_{rms}^2 \times R = \frac{I_0^2}{2}R$$

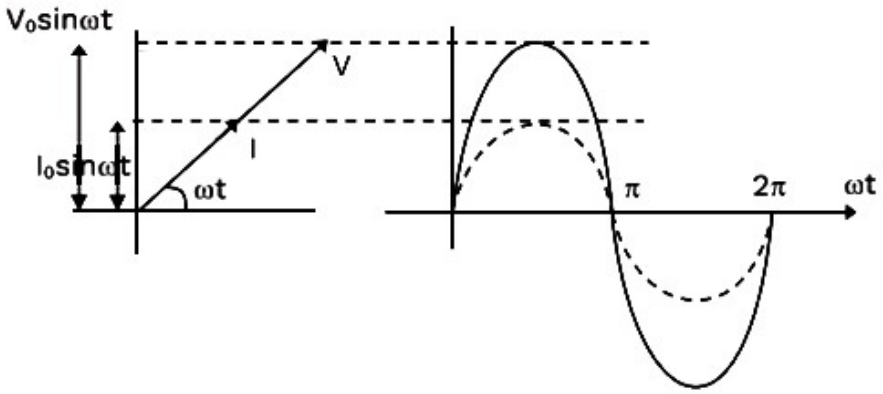
This effective value is defined as the root mean square value of current.

So, $I_{rms} = \sqrt{\frac{1}{2} I_0^2} = \frac{I_0}{\sqrt{2}}$

Similarly, $V_{rms} = \frac{V_0}{\sqrt{2}} = 0.707\ V_0$

Alternating voltage/current representation by phasors

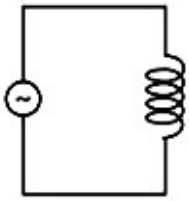
We use a phasor diagram to show the phase relationship between current and voltage in an a.c. circuit.
Phasors are vectors rotating about origin with angular speed ω. The peak values of voltage or current are represented by the length of the phasor.
Figure below shows the phasor diagram for a.c. applied to a resistor.



ALTERNATING VOLTAGE CONNECTED TO AN INDUCTOR

By Kirchhoff's voltage law, $\sum E(t) = 0$

So, applied voltage = Self-induced emf across inductor



Ideal inductor
L (with zero resistance)

$$V = \frac{LdI}{dt}$$

$$\frac{dI}{dt} = \frac{V}{L} = \frac{V_0 \sin\omega t}{L}$$

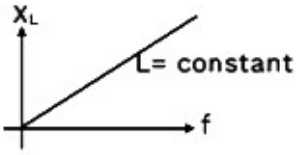
$$\int dI = \frac{V_0}{L}\int \sin\omega t\, dt$$

$$I = -\frac{V_0}{\omega L}\cos\omega t + C$$

Current is oscillating and it has no constant term, so C = 0.

$$\therefore \quad I = \frac{V_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right) = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$$

where $I_0 = \frac{V_0}{\omega L} = \frac{V_0}{X_L}$, and X_L = Resistance shown by inductor to a.c. flow. X_L is called inductive reactance.



(i) $X_L = \omega L = 2\pi fL$. On keeping L constant, X_L varies linearly with source frequency.
(ii) Keeping source frequency f constant, X_L varies linearly with self inductance L.

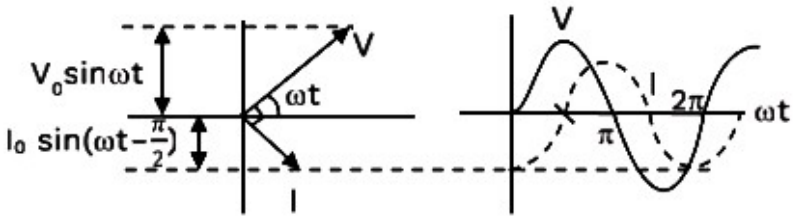
From the current and voltage equation, we can see that current through inductor lags behind the voltage by $\frac{\pi}{2}$. The instantaneous power supplied to inductor

$$P_L = I \times V$$

$$= I_0 \sin\left(\omega t - \frac{\pi}{2}\right).V_0 \sin\omega t$$

$$= -I_0 V_0 \sin\omega t \cos\omega t$$

$$P_L = -\frac{I_0 V_0}{2}\sin 2\omega t$$



Average power over a complete cycle

$\bar{P}_L = 0$ ..... (as < sin 2ωt > = 0)

Q. **For a given frequency, the inductive reactance of a coil at 100 H is 25 Ω. For the same frequency, the inductive reactance at 200 H will be?**

Sol. $X_L = 25\Omega,\ L = 100\,H$

$X_L' = ?,\ L' = 200\,H$

Since, $X_L \propto L$

$$\therefore \quad \frac{X_L}{X_L'} = \frac{L}{L'} \quad \text{or} \quad X_L' = X_L\left(\frac{L'}{L}\right) = 25\left(\frac{200}{100}\right) = 50\,\Omega$$

Q. **A pure inductor of 30 mH is connected to an AC source of 220 V, 50 Hz. The rms current in the inductor is**

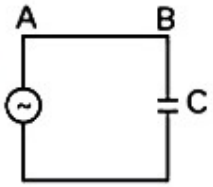
Sol. Given that, $L = 30\,mH = 30 \times 10^{-3}H;\quad V_{rms} = 220\,V$

Frequency, f = 50 Hz

Inductive reactance, $X_L = 2\pi f L = \left(2 \times 3.14 \times 50 \times 30 \times 10^{-3}\right) = 9.42\,\Omega$

Rms current, $I_{rms} = \frac{V_{rms}}{X_L} = \frac{220\,V}{9.42\,\Omega} = 23.34\,A$

ALTERNATING VOLTAGE CONNECTED TO A CAPACITOR



When a capacitor is connected to a.c., it gets charged and discharged in every cycle alternately.

If the charge on the capacitor at any instant is q, then

instantaneous voltage V across capacitor, $V = \frac{q}{C}$

By KVL, $\frac{q}{C} = V_0 \sin\omega t$

$$q = C.V_0 . \sin\omega t$$

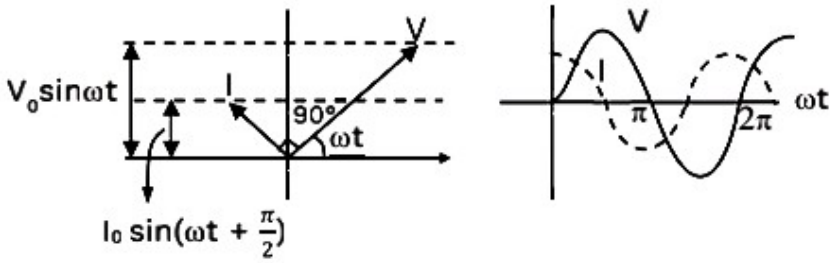
Current $I = \frac{dq}{dt} = CV_0 \times \omega \cos\omega t = \frac{V_0}{X_c} \cdot \sin\left(\omega t + \frac{\pi}{2}\right)$

where $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$ = Capacitive reactance

X_c is the resistance offered by the capacitor to the a.c. flow.

From the equation of current and voltage, we can see that current leads the voltage by $\frac{\pi}{2}$ for a capacitive circuit.

We can draw the phasor diagram as-



Instantaneous power, $P_C = V \times I = V_0 \sin\omega t . I_0 \sin\left(\omega t + \frac{\pi}{2}\right) = V_0 . I_0 . \sin\omega t \cos\omega t$

$$P_c = \frac{V_0 . I_0}{2} \sin 2\omega t$$

This equation shows that frequency of power is two times the source frequency.

Average power supplied to capacitor, over a complete cycle,

$$\bar{P}_C = \frac{V_0 . I_0}{2} < \sin 2\omega t > = 0$$

Q. **A capacitor of capacity C has reactance X. If the capacitance and frequency become double, then reactance will be?**

Sol. $X_C = \frac{1}{\omega C} = \frac{1}{(2\pi f)C}$

So, $\frac{X_{C_2}}{X_{C_1}} = \frac{f_1 C_1}{f_2 C_2} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$\therefore$ $X_{C_2} = \frac{X_{C_1}}{4} = \frac{X}{4}$

Q. **A capacitor of capacitance 10 μF is connected to an AC source and an AC ammeter. If the source voltage varies as $V = 50\sqrt{2}\sin 100t$, the reading of the ammeter is**

Sol. $C = 10\,\mu F = 10^{-5}\,F, \quad V = 50\sqrt{2}\sin 100t$

$\omega = 100\,rad\,s^{-1}$ and $V_0 = 50\sqrt{2}\,V$

$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 10^{-5}} = 1000\,\Omega$

$I_{rms} = \frac{V_{rms}}{X_C} = \frac{50\sqrt{2}}{\sqrt{2} \times 1000} = 0.05A = 50mA$

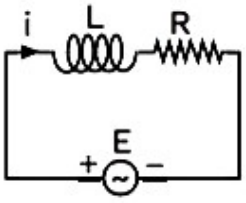
Alternating Current through an L-R Circuit

Let E = instantaneous emf,

And current, $i = I_0 \sin \omega t$

The voltage drop across the inductor, $V_L = \frac{Ldi}{dt}$

$\therefore \quad V_L = I_0 \omega L \cos \omega t = I_0 X_L \cos \omega t$



The voltage drop across the resistor, $V_R = iR$

$\therefore \quad V_R = I_0 R \sin \omega t$

Now, Applied voltage, $E = V_L + V_R$

$\Rightarrow \quad E = I_0 X_L \cos \omega t + I_0 R \sin \omega t$

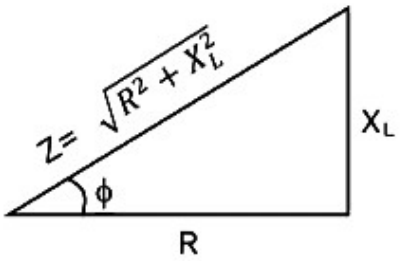
Let $X_L = Z \sin\phi$

$R = Z \cos\phi$

$\Rightarrow \quad E = I_0 Z \sin(\omega t + \phi)$ such that,

(a) $Z = \sqrt{R^2 + X_L^2}$

(b) $\tan\phi = \frac{X_L}{R} \Rightarrow$



$\Rightarrow \quad E = E_0 \sin(\omega t + \phi) \qquad (\because I_0 Z = E_0)$

Here, $E = E_0 \sin(\omega t + \phi)$

And, $I = I_0 \sin \omega t$

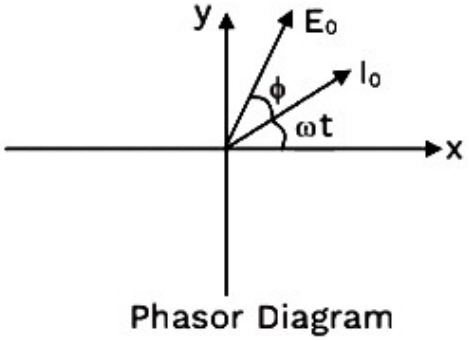
$\therefore$ Voltage leads current by a phase difference ϕ.

Also, $E_0 = I_0 Z$

or $\quad Z = \frac{E_0}{I_0} (\Omega)$

Z is called impedance of L-R circuit.

$Z = \sqrt{R^2 + X_L^2}$



Phasor Diagram

Alternating Current through an R-C Circuit

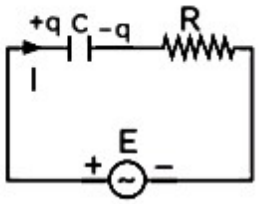
Let E= instantaneous emf, and,

Current at any instant, $I = I_0 \sin \omega t$.

Potential difference across capacitor, $V_C = I_0 X_C \sin(\omega t - \pi/2)$

Potential difference across resistor, $V_R = I_0 R \sin(\omega t)$

$\therefore$ Applied voltage, $E = V_C + V_R$

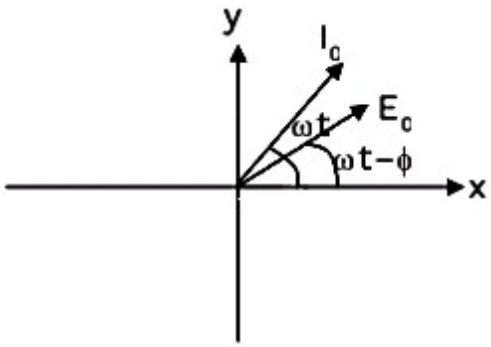


or $E = I_0 [-X_C \cos\omega t + R \sin\omega t]$

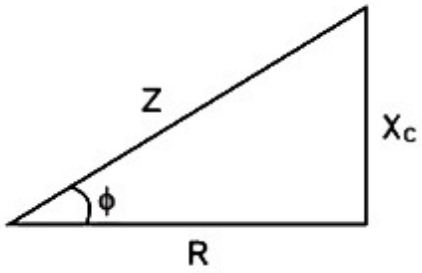
Let $R = Z\cos\phi$, $X_C = Z\sin\phi$

$\therefore \quad Z = \sqrt{R^2 + X_C^2}$, $\tan\phi = \dfrac{X_C}{R}$

$\Rightarrow \quad E = I_0 Z [\sin\omega t \cos\phi - \cos\omega t \sin\phi]$



Phasor Diagram



Impedance Triangle

$E = I_0 Z \sin(\omega t - \phi)$

$\therefore$ Voltage lags behind current by a phase difference ϕ.

$Z = \sqrt{R^2 + X_C^2}$ = Impedance

POWER IN AC CIRCUITS

At any instant, $V = V_0 \sin\omega t$

And, $I = \dfrac{V_0}{Z} \sin(\omega t + \phi) = I_0 \sin(\omega t + \phi)$

$P = V \times I = V_0 . I_0 . \sin\omega t . \sin(\omega t + \phi)$

$$= \frac{V_0 . I_0}{2} \cdot [\cos(\omega t + \phi - \omega t) - \cos(\omega t + \omega t + \phi)]$$

$$P = \frac{V_0 I_0}{2}\cdot[\cos\phi - \cos(2\omega t + \phi)]$$

Now, $\cos(2\omega t + \phi) = 0$ over one cycle.

So, $\bar{P} = V_{rms} I_{rms} \cos\phi$

$$\bar{P} = \frac{V_{rms}^2}{Z}\cdot\cos\phi$$

$$\bar{P} = I_{rms}^2 \times Z \times \cos\phi = I_{rms}^2 \times Z \times \frac{R}{Z} = I_{rms}^2 \times R$$

(From impedance triangle, $\cos\phi = \frac{R}{Z}$)

$\cos\phi$ is called power factor as it is used to find power consumed in an a.c. circuit.

In pure resistive circuit, phase difference between voltage and current is zero, i.e., $\phi = 0°$,

Then, $\cos\phi = 1$, so the power dissipated in pure resistive a.c. circuit is maximum.

$P = V_{rms} I_{rms}$

In Pure inductive or pure Capacitive circuits, the phase difference between current and voltage is $\phi = \frac{\pi}{2}$.

So, the power factor $\cos\phi = 0$.

Thus, the power consumed in pure inductive or pure capacitive a.c. circuit is zero.

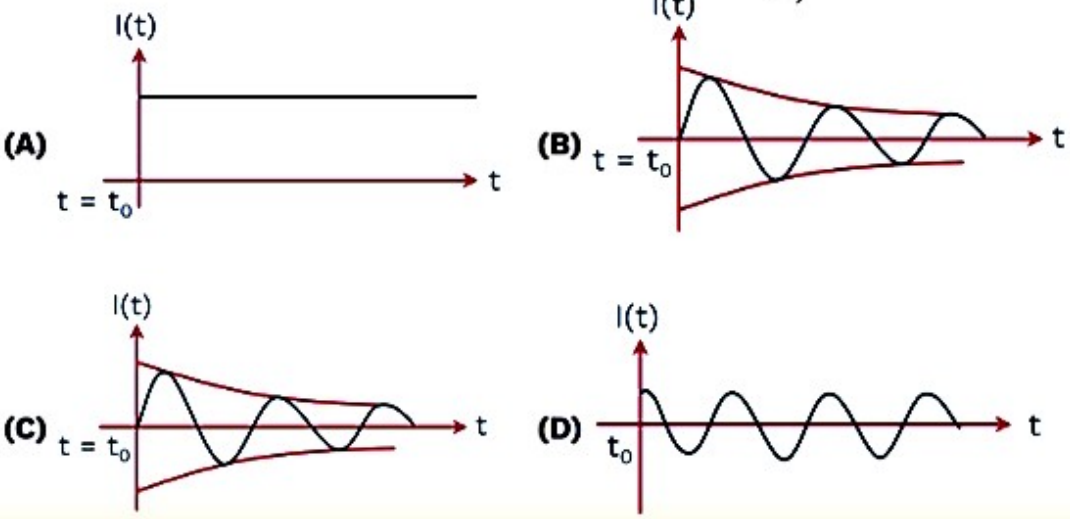
Such a current which causes no power consumption in the circuit is called "wattless current".

Q. **An electric motor when loaded has an effective resistance of 30 Ω and an inductive reactance of 40 Ω. If the motor is powered by a source with maximum voltage of 420 V, the maximum current is**

Sol. $R = 30\,\Omega, X_L = 40\,\Omega, V = 420\,V$

$$Z = \sqrt{R^2 + X_L^2} = 50\,\Omega \Rightarrow I = \frac{V}{Z} = \frac{420}{50} A = 8.4\,A$$

Q. **A series LR circuit is connected to a voltage source with $V(t) = V_0 \sin\omega t$. After very large time, current I(t) behaves as $\left(t_0 >> \frac{L}{R}\right)$**



Sol. Current in LR circuit is

$$I = \frac{V_0}{\sqrt{R^2 + \omega^2 L^2}} \sin\left(\omega t - \frac{\pi}{2}\right)$$

i.e., it is sinusoidal in nature with phase lag of $\frac{\pi}{2}$. So, option D is correct.

Q. **A series combination of resistor (R), capacitor (C) is connected to an AC source of angular frequency ω. Keeping the voltage same, if the frequency is changed to $\frac{\omega}{3}$, the current becomes half of the original current. Then the ratio of the capacitive reactance and resistance at the former frequency is**

Sol.

For case 1, capacitive reactance, $X_c = \frac{1}{\omega C}$

Impedance of the circuit, $Z = \sqrt{R^2 + X_c^2}$

Current in the circuit, $I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + X_c^2}}$ ...(i)

For case 2, $\omega' = \frac{\omega}{3}$

$\therefore$ Capacitive reactance, $X_c' = \frac{1}{\omega' C} = \frac{1}{\frac{\omega}{3}C} = \frac{3}{\omega C} = 3X_c$

Impedance of the circuit, $Z' = \sqrt{R^2 + X_c'^2} = \sqrt{R^2 + 9X_c^2}$

Current in the circuit, $\frac{I}{2} = \frac{V}{Z'} = \frac{V}{\sqrt{R^2 + 9X_c^2}}$ ...(ii)

Divide (ii) by (i), we get

$$\frac{1}{2} = \frac{\sqrt{R^2 + X_c^2}}{\sqrt{R^2 + 9X_c^2}} \Rightarrow \frac{1}{4} = \frac{R^2 + X_c^2}{R^2 + 9X_c^2}$$

$$5X_c^2 = 3R^2 \text{ or } \frac{X_c}{R} = \sqrt{\frac{3}{5}}$$

Q. **When 100 volt dc is applied across a coil, a current of 1 A flows through it, when 100 volt ac of 50 Hz is applied to the same coil, only 0.5 A flows. Calculate the resistance and inductance of the coil.**

Sol. The coil acts as a combination of inductor and resistor. For given L-R circuit,

$$I = \frac{V}{Z}, Z = \sqrt{R^2 + (\omega L)^2}$$

So when dc is applied, $\omega = 0$, $Z = R$

And hence $R = 100\Omega$

And when ac of 50 Hz is applied

$$I = \frac{V}{Z}, \quad Z = \frac{V}{I} = \frac{100}{0.5} = 200\,\Omega$$

But $Z = \sqrt{R^2 + (\omega L)^2}$

$(2\pi fL)^2 = 200^2 - 100^2 = 3 \times 10^4$

$$L = \frac{\sqrt{3} \times 10^2}{2\pi \times 50} = \frac{\sqrt{3}}{\pi} H = 0.55 H$$

Q. **In an L-R series circuit $\left(L = \frac{175}{11} mH \text{ and } R = 12\Omega\right)$, a variable emf source ($V = V_0 \sin\omega t$) of $V_{rms} = 130\sqrt{2}V$ and frequency 50 Hz is applied. The current amplitude in the circuit and phase of current with respect to voltage are respectively $\left(\text{use } \pi = \frac{22}{7}\right)$.**

Sol. Given $L = \frac{175}{11} mH$ and $R = 12\Omega$

$\omega = 2\pi f = 100\pi$

Impedance in a L–R Circuit, $Z = \sqrt{(\omega L)^2 + R^2} = 13\Omega$

Voltage amplitude $V_0 = V_{rms} \times \sqrt{2} \quad \Rightarrow V_0 = (130 \times 2) = 260\,V$

Current amplitude $I_0 = \frac{V_0}{Z} = \frac{130 \times 2}{13} = 20A$

Phase angle $\phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \tan^{-1}\left(\frac{5}{12}\right)$

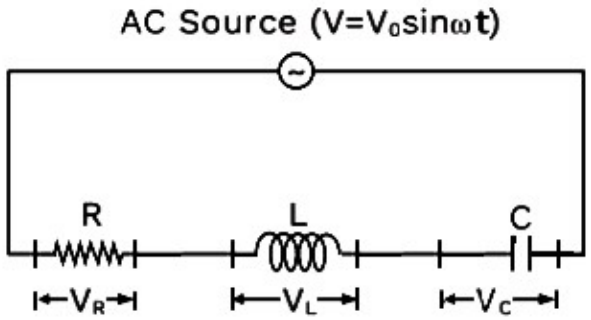
ALTERNATING VOLTAGE CONNECTED TO A SERIES LCR CIRCUIT

In a series connection of capacitor C, inductor L and resistor R across an alternating voltage, let q be the instantaneous charge on the capacitor and I be the instantaneous current in the circuit. Then, by KVL,

$$IR + L\frac{dI}{dt} + \frac{q}{C} = V$$

Same current flows through all the elements in series.

So, $I = I_R = I_L = I_C$



Voltage across the ends of resistor, $V_R = I \times R$

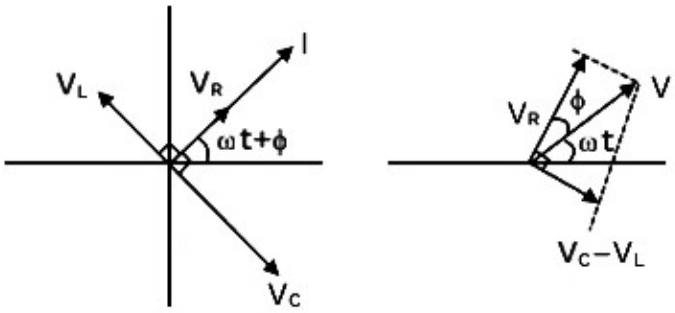
Voltage across the ends of inductor,

$$V_L = IX_L = I\omega L = I \times 2\pi f \times L$$

Voltage across the ends of capacitor,

$$V_C = IX_C = \frac{I}{\omega C} = \frac{I}{2\pi fC}$$

V_R and I are in same phase, V_L leads I by a phase $\frac{\pi}{2}$ and V_C lags I by a phase $\frac{\pi}{2}$.



As a result, V_L and V_C have a phase difference of π.

Voltage across the combination of the inductor and capacitor is then equal to the magnitude of $V_C - V_L$.

The total voltage then becomes,

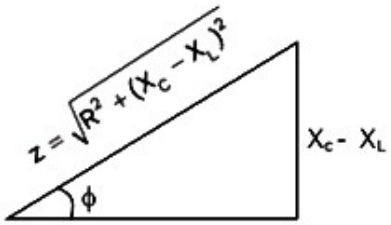
$$V = \sqrt{V_R^2 + (V_C - V_L)^2}$$

$$V = \sqrt{(IR)^2 + (IX_C - IX_L)^2} = I\sqrt{R^2 + (X_C - X_L)^2}$$

$$\frac{V}{I} = \sqrt{R^2 + (X_C - X_L)^2} = Z \text{ (impedance = Effective resistance of series LCR circuit)}$$

Phase difference between V and I

$$\tan\phi = \frac{V_C - V_L}{V_R} = \frac{X_C - X_L}{R}$$



Impedance Triangle

Q. **A sinusoidal voltage of peak value 283 V angular frequency 320 rad/s is applied to a series LCR circuit. Given that R = 5Ω, L = 25mH and C = 100μF . The total impedance, and phase difference between the voltage across the source and the current will respectively be**

(A) 10 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

(B) 10 Ω and $\tan^{-1}\left(\frac{8}{3}\right)$

(C) 7 Ω and $\tan^{-1}\left(\frac{5}{3}\right)$

(D) 7 Ω and 45°

Sol. Here, $E_0 = 283V$, $\omega = 320\ \text{rad s}^{-1}$, $R = 5\Omega$,

$L = 25\text{mH} = 25 \times 10^{-3}\text{H}$,

$C = 1000\,\mu\text{F} = 10^{-3}\text{F}$

$X_L = \omega L = 320 \times 25 \times 10^{-3} = 8\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{320 \times 10^{-3}} = \frac{1000}{320} = 3.125\,\Omega$$

Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{5^2 + (8 - 3.125)^2} \approx \sqrt{49} = 7\Omega$

Phase difference, $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$

$\phi = \tan^{-1}\left(\frac{4.875}{5}\right) \approx \tan^{-1}(1) = 45^\circ$

Q. **An LCR series circuit with 100 Ω resistance is connected to an ac source of 200V and angular frequency 300 rad/s. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Calculate the current and the power dissipated in the LCR circuit.**

Sol. When capacitor is removed,

$$\Rightarrow \quad \tan 60 = \frac{X_L}{R}$$

$$\Rightarrow \quad X_L = \sqrt{3}R$$

When inductor is removed,

$$\Rightarrow \quad \tan 60 = \frac{X_C}{R}$$

$$\Rightarrow \quad X_C = \sqrt{3}R$$

Hence $X_C = X_L$

Then the impedance is given by

$$\Rightarrow \quad Z = \sqrt{R^2 + (X_L - X_C)^2} = R = 100$$

Then the current is

$$\Rightarrow \quad I = \frac{E}{Z} = \frac{200}{100} = 2A$$

Then the power is given by

$$\Rightarrow \quad P = EI\cos\phi = EI\frac{R}{Z}$$

$$\Rightarrow \quad P = (200) \times 2 \times \frac{100}{100} = 400W$$

Q. **An alternating voltage of 220 volt r.m.s at a frequency of 40 cycle/sec is supplied in a circuit containing a pure inductance of 0.01 H and a pure resistance of 6 ohms in series. Calculate (i) the current, (ii) potential difference across the resistance, (iii) potential difference across the inductor, (iv) the time tag, (v) power factor**

Sol. Given, $V_{rms} = 220V$, $f = 40$ Hz $\Rightarrow \omega = 80\pi$

and $L = 0.01$ H $\Rightarrow X_L = \omega L = 0.8\pi = 2.512\Omega$

$R = 6\Omega$

$\Rightarrow$ Impedence, $Z = \sqrt{R^2 + X_L^2} = \sqrt{36 + 6.3} = \sqrt{43.3} = 6.58$

So, rms current, $I_{rms} = \frac{V}{Z} = \frac{220}{6.58} = 33.43A$

Potential difference across the resistance, $V_R = I_{rms} R = 33.43 \times 6 = 200.58$ V

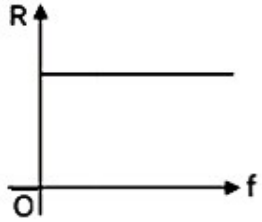
Potential difference across the inductor is $V_L = I_{rms} X_L = 33.43 \times 2.512 = 83.97V$

For time lag, $\omega t = \tan^{-1} \frac{V_L}{V_R} = 22.7$

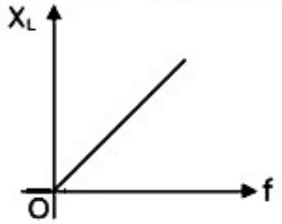
Power factor, $\cos \phi = \cos 22.7 = 0.922$

Impedance variation with source frequency

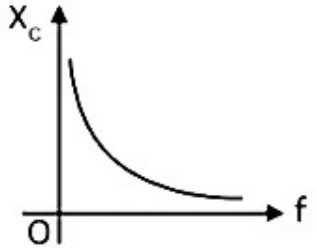
(i) Resistance R does not depend upon source frequency.



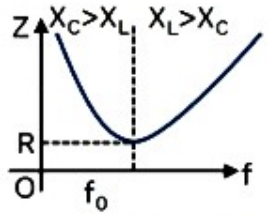
(ii) $X_L = \omega L = 2\pi fL$, so X_L varies linearly with source frequency.



(iii) $X_C = \dfrac{1}{\omega C} = \dfrac{1}{2\pi fC}$, X_C varies inversely with source frequency.



(iv) Impedance, $Z = \sqrt{R^2 + (X_C - X_L)^2}$



Impedance is minimum when $X_C = X_L$.

Resonance

For $X_C = X_L$, which gives $\dfrac{1}{\omega C} = \omega L \quad \Rightarrow \boxed{f = \dfrac{1}{2\pi\sqrt{LC}} = f_0}$

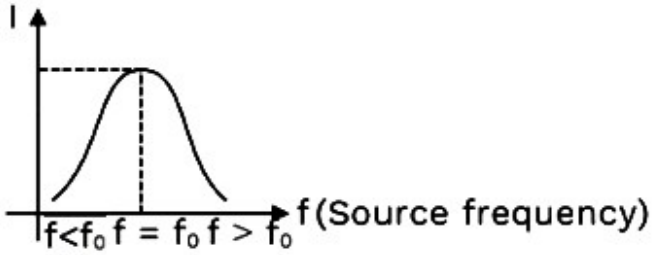
Here, f_0 is a is called **resonant frequency**.

At resonance, impedance, $Z = Z_{min} = R$

At resonant frequency, the circuit is purely resistive and the circuit impedance is minimum. Also, the voltage drop across capacitor and inductor is zero, so net voltage is equal to the voltage across resistor.

The current $I = \frac{V}{Z}$ is maximum when impedance Z is minimum.

$$I_{max} = \frac{V}{Z_{min}} = \frac{V}{R}$$



Q. **In a series LCR circuit the voltages across resistance, capacitance and inductance are 20 V each. If the capacitance is short-circuited, the voltage across the inductance will be**

Sol. For a series LCR circuit,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} \quad \text{Given that } V_R = V_L = V_C = 20V$$

$\therefore$ $V = \sqrt{(20V)^2} = 20V$ and $R = X_L = X_C$

When the capacitor is short circuited, the impedance of the circuit becomes

$$Z' = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + R^2} = \sqrt{2}R$$

and the current in the circuit, $I' = \frac{V}{Z'} = \frac{20V}{\sqrt{2}R}$

$\therefore$ The voltage across the inductance is

$$V_L' = I' X_L = \left(\frac{20V}{\sqrt{2}R}\right) R = \frac{20}{\sqrt{2}} V$$

Q. **An LCR circuit contains R = 50 Ω, L = 1mg and C = 0.1μF. The impedance of the circuit will be minimum for a frequency**

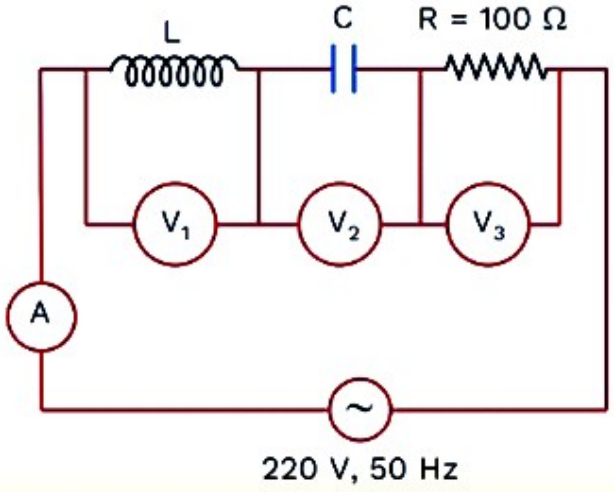
Sol. The impedance of the LCR circuit will be minimum at resonant frequency f_0.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Here, $L = 1mH = 1\times10^{-3}H$, $C = 0.1\mu F = 0.1\times10^{-6}F$

$$\therefore \quad f_0 = \frac{1}{2\pi\sqrt{\left(1\times10^{-3}H\right)\left(0.1\times10^{-6}F\right)}} = \frac{10^5}{2\pi}Hz$$

Q. **In the given circuit, if the reading of voltmeters V_1 and V_2 are 300 volts each, then the reading of the voltmeter V3 and ammeter A are**



Sol. As $V = \sqrt{V_R^2 + \left(V_L - V_C\right)^2} = V_R \left(\text{As } V_L = V_C = 300V\right)$

$$\therefore \quad V_R = 220V = V_3 \text{ and } I = \frac{V_R}{R} = \frac{220V}{100\Omega} = 2.2A$$

Q. **When a DC voltage of 200 V is applied to a coil of self inductance $\left(\frac{2\sqrt{3}}{\pi}\right)$H, a current of 1 A flows through it. But by replacing DC source with AC source of 200 V, the current in the coil is reduced to 0.5 A. Then the frequency of AC supply is**

Sol. As current flows in the coil when it is connected to DC voltage, so there must be some resistance in the coil.

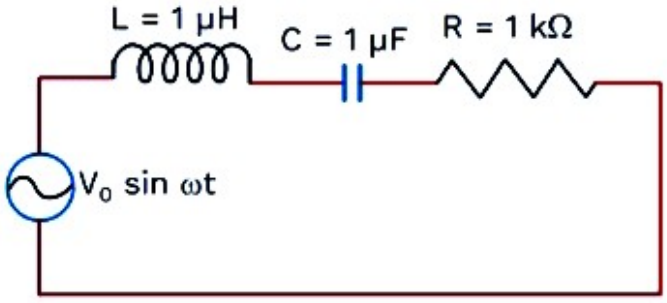
Resistance of coil $R = \frac{200V}{1A} = 200\Omega$

With AC source, $I = \frac{200}{\sqrt{R^2 + X_L^2}}$ or $0.5 = \frac{200}{\sqrt{R^2 + X_L^2}}$

or $R^2 + (2\pi fL)^2 = (400)^2$

or $\left(2\pi f \times \frac{2\sqrt{3}}{\pi}\right)^2 = (400)^2 - (200)^2 \Rightarrow f^2 = \frac{12 \times 10^4}{16 \times 3} = 2500 \Rightarrow f = 50Hz$

Q. **In the circuit shown, L = 1μH, C = 1μF and R = 1kΩ. They are connected in series with an a.c. source $V = V_0 \sin\omega t$ as shown. Which of the following options is/are correct?**



(A) At ω – 0, the current flowing through the circuit becomes nearly zero.
(B) The frequency at which the current will be in phase with the voltage is independent of R.
(C) The current will be in phase with the voltage if $\omega = 10^6$ rad s^{-1}.
(D) At $\omega >> 10^6$ $rads^{-1}$, the circuit behaves like a capacitor.

Sol. At $\omega \sim 0$, the impedance approaches infinity and current is nearly zero. A is correct.

At resonant frequency, current and voltage will be in same phase i.e.,

$$\omega L = \frac{1}{\omega C}$$

$$\therefore \quad \text{Resonant frequency } \omega = \frac{1}{\sqrt{LC}}$$

Clearly, frequency is independent of R. B is correct.

Here, $L = 1\mu H = 10^{-6} H$, $C = 10^{-6} F$

$$\therefore \quad \omega = \frac{1}{\sqrt{10^{-12}}} = 10^{6} \text{rads}^{-1}$$. C is correct.

Since, $X_C = \frac{1}{\omega C}$, $X_L = \omega L$

For large ω, $X_C \sim 0$, so circuit will behave like a inductive circuit. D is incorrect.

Sharpness of Resonance

The current amplitude in series LCR circuit is given by

$$I_0 = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

and is maximum when $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$.

The maximum value is, $(I_0)_{max} = \frac{V_0}{R}$.

The values of ω where the current is $\frac{1}{\sqrt{2}}$ times its maximum value are called the half power frequencies because at these frequencies, the power dissipated in the circuit becomes half. There are two such values of ω, lets say, ω_1 and ω_2, one greater and the other smaller than ω_0 and symmetrical about ω_0. Then,

$\omega_1 = \omega_0 + \Delta\omega$

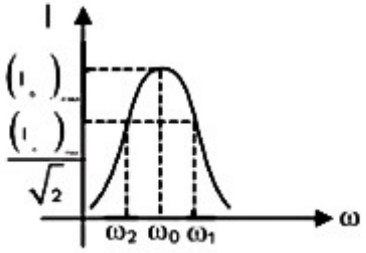
$\omega_2 = \omega_0 + \Delta\omega$

The difference $\omega_1 - \omega_2 = 2\Delta\omega$ is known as **bandwidth** of the circuit.

The quantity $\left(\frac{\omega_0}{2\Delta\omega}\right)$ is a measure of the sharpness of resonance.

Smaller the value of $\Delta\omega$, the sharper or narrower is the resonance.
Therefore,

At ω_1, $I_0 = \frac{V_0}{\sqrt{R^2 + \left(\omega_1 L - \frac{1}{\omega_1 C}\right)^2}}$



Also, $\frac{(I_0)_{max}}{\sqrt{2}} = \frac{V_0}{R\sqrt{2}}$

So, $\sqrt{R^2 + \left(\omega_1 L - \frac{1}{\omega_1 C}\right)^2} = R\sqrt{2}$

or $R^2 + \left(\omega_1 L - \frac{1}{\omega_1 L}\right)^2 = 2R^2$

$$\omega_1 L - \frac{1}{\omega_1 C} = R$$

We can rewrite the above equation as,

$$\left(\omega_0 + \Delta\omega\right)L - \frac{1}{\left(\omega_0 + \Delta\omega\right)C} = R$$

$$\omega_0 L\left(1 + \frac{\Delta\omega}{\omega_0}\right) - \frac{1}{\omega_0 C\left(1 + \frac{\Delta\omega}{\omega_0}\right)} = R$$

By using $\omega_0^2 = \frac{1}{LC}$, we can write $\frac{1}{\omega_0 C} = \omega_0 L$, so we get,

$$\omega_0 L\left(1 + \frac{\Delta\omega}{\omega_0}\right) - \frac{\omega_0 L}{\left(1 + \frac{\Delta\omega}{\omega_0}\right)} = R$$

As $\frac{\Delta\omega}{\omega_0} << 1$, so by binomial approximation, $\left(1 + \frac{\Delta\omega}{\omega_0}\right)^{-1}$ as $\left(1 - \frac{\Delta\omega}{\omega_0}\right)$

So,

$$\omega_0 L\left(1+\frac{\Delta\omega}{\omega_0}\right)-\omega_0 L\left(1-\frac{\Delta\omega}{\omega_0}\right)=R$$

or $\omega_0 L\dfrac{2\Delta\omega}{\omega_0}=R$

$$\boxed{\Delta\omega=\frac{R}{2L}}$$

The sharpness of resonance is given by,

$$\frac{\omega_0}{2\Delta\omega}=\frac{\omega_0 L}{R}=Q$$

The ratio $\dfrac{\omega_0 L}{R}$ is known as quality factor, Q of the circuit.

We can write, $2\Delta\omega=\dfrac{\omega_0}{Q}$.

Greater the quality factor, smaller is the bandwidth and hence the resonance is sharper.

Parallel Resonance Circuit

In a parallel resonance RLC circuit, $X_L = X_C$.

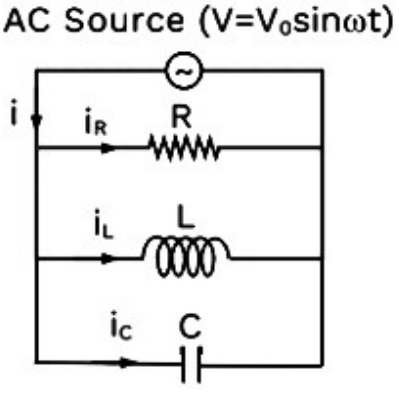
(i) $\dfrac{1}{Z}=\sqrt{\dfrac{1}{R^2}+\left(\dfrac{1}{X_C}-\dfrac{1}{X_L}\right)^2} \Rightarrow Z=R$

(ii) $i_L=\dfrac{V_0}{X_L}\sin\left(\omega t-\dfrac{\pi}{2}\right), \quad i_C=\dfrac{V_0}{X_C}\sin\left(\omega t+\dfrac{\pi}{2}\right)$

$\Rightarrow \quad i_L=-\dfrac{V_0}{X_L}\cos\omega t, \quad i_C=\dfrac{V_0}{X_C}\cos\omega t$

$\Rightarrow \quad i_L+i_C=0$

$\Rightarrow \quad i=i_R=\dfrac{V_0}{R}\sin\omega t$



(iii) $\tan\phi=\dfrac{\dfrac{1}{X_C}-\dfrac{1}{X_L}}{R}=0$.

∴ Current and emf are in same phase.

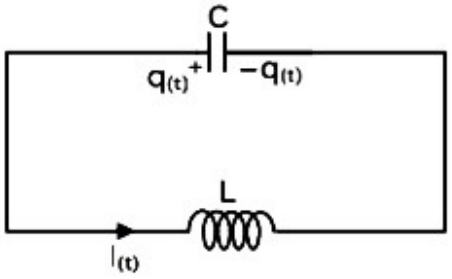
LC Oscillator Circuit

Let a capacitance C having initial charge q_0 be connected to an inductor at t = 0.

After time t, consider the charge on the capacitor to be qt and the current across the inductor be It .

Then, by KVL,

$$V_C = V_L \quad \Rightarrow \frac{q}{C} = L\frac{dI}{dt}$$



The current flow across the inductor is a result of loss of charge by the capacitor, so we can write, $I = \frac{-dq}{dt}$

Then,

$$\frac{q}{C} = L\frac{d}{dt}\left[\frac{-dq}{dt}\right] = -L\frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} = \frac{-1}{LC} \times q$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

This is a 2nd order differential equation having similar form as that of SHM equation

So, we can say that the charge oscillates in this circuit with frequency $\omega_0 = \frac{1}{\sqrt{LC}}$

The solution of a 2nd order differential equation is given by,

$q(t) = q_0 \cos(\omega_0 t + \phi)$

Using initial condition, at t = 0, $q = q_0$

$\Rightarrow \quad \cos\phi = 1$ or $\phi = 0$

So, $q_t = q_0 \cos(\omega_0 t)$

The current $i\left(= -\frac{dq}{dt}\right)$ is then,

$$i = i_0 \sin(\omega_0 t) \qquad \text{where } i_0 = \omega_0 q_0$$

From above equations, it can be said that current and charge in the circuit have a sinusoidal oscillation. It is known as LC oscillation.

Similar to SHM in mechanics, the energy of the inductor and the capacitor also oscillates while the total energy of the circuit remains constant.

Initially, electrical energy is stored in the charged capacitor. $U_E = \frac{1}{2}\frac{q_0^2}{C}$.

Since, there is no current in the circuit, energy in the inductor is zero. Thus, the total energy of LC circuit is, $U = U_E = \frac{1}{2}\frac{q_0^2}{C}$

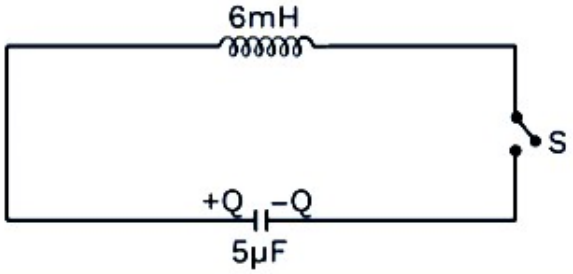
On closing the switch, capacitor starts discharging and it produces some magnetic field in the inductor. As a result, some magnetic energy is stored in the inductor, $U_B = \left(\frac{1}{2}\right)Li^2$.

As the current becomes maximum (i_0) at $t = \frac{T}{4}$, all the energy is stored in the magnetic field: $U_B = \left(\frac{1}{2}\right)Li_0^2$. and the energy in the capacitor becomes zero.

The current now starts charging the capacitor. This process continues till the capacitor is fully charged at $t = \frac{T}{2}$. But the polarity of the capacitor is reversed.

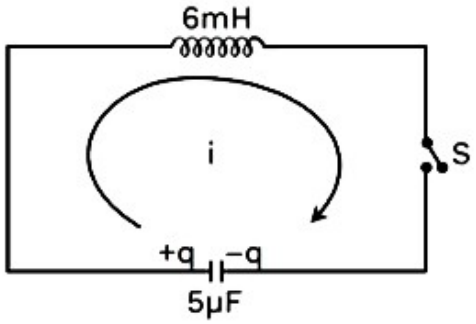
The whole process repeats till the system reaches its original state. Thus, the energy in the system oscillates between the capacitor and the inductor.

Q. **In the circuit shown in figure, charge on capacitor is Q = 100μC. If switch is closed at t = 0, find the current in circuit when charge on capacitor reduces to 50μC. Also find the maximum current in circuit.**



Sol. At t = 0 when switch is closed, capacitor starts discharging through the inductor and current in circuit increases. When charge on capacitor is q and current in circuit is I, then by conservation of energy,

$$\frac{q^2}{2C} + \frac{1}{2}Li^2 = \frac{Q^2}{2C}$$



Current in circuit in terms of charge on capacitor is given as

$$i = \frac{1}{\sqrt{LC}}\sqrt{Q^2 - q^2} = \frac{\sqrt{(100\mu C)^2 - (50\mu C)^2}}{\sqrt{3 \times 10^{-8}}}$$

$$\Rightarrow \quad i = \frac{10^{-6}}{\sqrt{3} \times 10^{-4}}\sqrt{7500}$$

$$\Rightarrow \quad i = 0.5A$$

Maximum energy stored in inductor is given as

$$\frac{1}{2}LI_{max}^2 = \frac{Q^2}{2C}$$

$$\Rightarrow \quad I_{max} = \frac{Q}{\sqrt{LC}} = \frac{100 \times 10^{-6}}{\sqrt{3 \times 10^{-8}}} = \frac{1}{\sqrt{3}}A$$

Q. **In an L-C circuit, L = 3.3H and C = 840pF. At t = 0, charge on the capacitor is 105μC and maximum. Compute the following quantities at t = 2.0 ms**
(a) The energy stored in the capacitor
(b) The total energy in the circuit
(c) The energy stored in the inductor

Sol. The angular frequency of L-C oscillations is,

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \quad \omega = \frac{1}{\sqrt{3.3 \times 840 \times 10^{-12}}}$$

$$\Rightarrow \quad \omega = 1.9 \times 10^4 \text{ rad/s}$$

Charge stored in the capacitor at time t, is given as

$$q = q_0 \cos \omega t$$

(a) At $t = 2 \times 10^{-3}$s

$$q = (105 \times 10^{-6}) \cos[1.9 \times 10^4] [2 \times 10^{-3}]$$

$$\Rightarrow \quad q = 100.3 \times 10^{-6} \text{ C}$$

Energy stored in the capacitor at this instant is given as

$$U_c = \frac{1}{2}\frac{q^2}{C}$$

$$\Rightarrow \quad U_c = \frac{\left(100.3 \times 10^{-6}\right)^2}{2 \times 840 \times 10^{-12}}$$

$$\Rightarrow \quad U_c = 6.0 \text{ J}$$

(b) Total energy in the circuit is the energy in capacitor at maximum charge which is given as

$$U = \frac{1}{2}\frac{q_0^2}{C}$$

$$\Rightarrow \quad U = \frac{\left(105 \times 10^{-6}\right)^2}{2 \times 840 \times 10^{-12}}$$

$$\Rightarrow \quad U = 6.56 \text{ J}$$

(c) Energy stored in inductor in the given time is given by conservation of energy as

$$U_L = (6.56 - 6.0)\text{J}$$

$$\Rightarrow \quad U_L = 0.56 \text{ J}$$

Q. **In an oscillating LC circuit in which C = 4.00μF, the maximum potential difference across the capacitor during the oscillations is 1.50 V and the maximum current through the inductor is 50.0mA.**
(a) What is the inductance L?
(b) What is the frequency of the oscillations?
(c) How much time does the charge on the capacitor take to rise from zero to its maximum value?

Sol. (a) By conservation of energy we have

$$\frac{1}{2}Li_0^2 = \frac{1}{2}CV_0^2L$$

$$\Rightarrow \quad L = \frac{CV_0^2}{i_0^2}$$

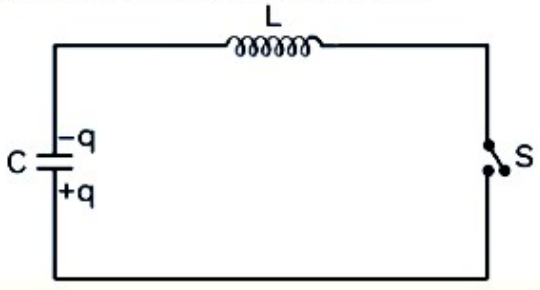
$$\Rightarrow \quad L = \frac{\left(4\times10^{-6}\right)(1.5)}{\left(50\times10^{-3}\right)^2}$$

$$\Rightarrow \quad L = 3.6\times10^{-3}\ H$$

(b) Oscillation frequency of the LC circuit is given as

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Q. **In figure. If S is closed at t = 0, find charge on capacitor as a function of time. The inductor is active with initial current I0 in it.**



Sol. If q_0 is the maximum charge on capacitor then by energy conservation we have

$$\frac{1}{2}LI_0^2 = \frac{q_0^2}{2C} \quad \Rightarrow \quad q_0 = \sqrt{LC}\cdot I_0$$

Charge on capacitor as an function of time is given as

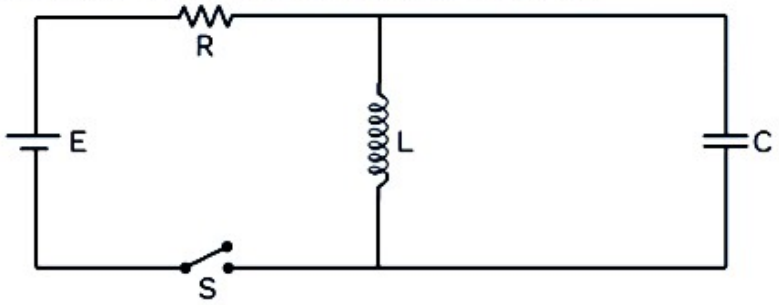
$$q = q_0 \sin(\omega t + \alpha)$$

Where $\omega = \frac{1}{\sqrt{LC}}$ and at t = 0, q = 0 we get $\alpha = 0$.

Thus charge on capacitor is given as

$$q = \sqrt{LC}\, I_0 \sin\left(\frac{t}{\sqrt{LC}}\right)$$

Q. **In the circuit shown in the figure, E = 50.0V, R = 250Ω and C = 0.500μF. The switch S is closed for a long time, and no voltage is measured across the capacitor. After the switch is opened, the voltage across the capacitor reaches a maximum value of 150 V. What is the inductance L?**



Sol. In steady state when switch was closed,

$$i_0 = \frac{E}{R} = 0.2A$$

After switch is opened, it becomes LC circuit in which peak current is 0.2A.

By energy conservation,

$$\frac{1}{2}Li_0^2 = \frac{1}{2}CV_0^2$$

$$\Rightarrow \quad L = \frac{V_0^2}{i_0^2} \cdot C$$

$$\Rightarrow \quad L = \frac{(150)^2}{(0.2)^2} \times 0.5 \times 10^{-6}$$

$$\Rightarrow \quad L = 0.28 \text{ H}$$

TRANSFORMERS

Transformers are electrical devices working on the principle of mutual induction. They are used to step up or step down the alternating voltages.

A simple transformer has two coils wrapped on an iron core. The coils are known as primary and secondary coils.

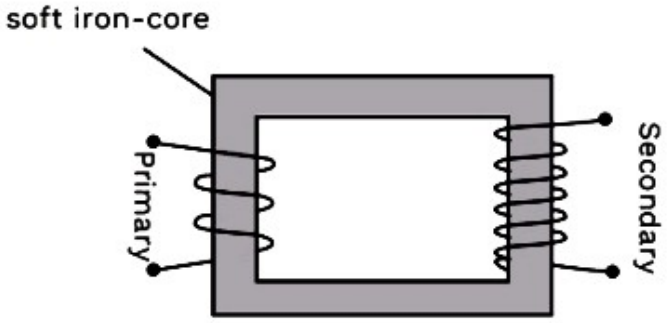
We apply the input a.c. voltage across the primary coil and the output a.c. voltage is taken across the secondary coil.

If N_p is the number of turns in the primary coil and N_s is the number of turns in the secondary coil, then

Induced emf across the primary coil, $E_p = -N_p \frac{d\phi}{dt}$

Similarly, induced emf across secondary coil, $E_s = -N_s \frac{d\phi}{dt}$

Now, $\frac{\text{AC voltage across secondary coil}}{\text{AC voltage across primary coil}} = \frac{V_s}{V_p} = \frac{E_s}{E_p} = \frac{N_s}{N_p}$



For a transformer with no energy loss,

$$P = I_p V_p = I_s V_s$$

So, $\frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$

So, $V_s = \left(\frac{N_s}{N_p}\right) V_p$ and $I_s = \left(\frac{N_p}{N_s}\right) I_p$

For step up transformer, $V_s > V_p$ so $N_s > N_p$ and $I_s < I_p$

For step down transformer, $V_s < V_p$ so $N_s < N_p$ and $I_s > I_p$

In actual transformers, there are small energy losses due to

(i) Flux leakage

(ii) Winding resistance

(iii) Eddy current in the core

(iv) Hysteresis losses

Transformers are used to step up the voltage for transmission of electricity over large distances. By stepping up the voltage, the output current reduces and the I2R losses in the transmission wires is minimised.

The voltage is stepped down at thed istributor power houses and then supplied to the homes.

Q. **An ideal transformer has 50 turns in its primary winding and 25 turns in its secondary winding. If the current in the secondary winding in 4A, what is the current in primary winding if a 200V AC is applied across it?**

Sol. With the given ratio of turns in primary to secondary winding it is a 2 : 1 step down transformer for which by transformer equation we have

$$\frac{e_s}{e_p} = \frac{N_s}{N_p}$$

$$\Rightarrow \quad e_s = \frac{N_s}{N_p} \times e_p$$

$$\Rightarrow \quad e_s = \frac{25}{50} \times 200 = 100V$$

Power equation of ideal transformer gives

$$e_s i_s = e_p i_p$$

$$\Rightarrow \quad i_p = \frac{e_s i_s}{e_p} = \frac{100 \times 4}{200} = 2A$$

Q. **An ideal power transformer is used to step up an alternating EMF of 220V to 4.4kV to transmit 0.6kW of power. If the primary coil has 1000 turns, find**
(a) The number of turns in the secondary coil
(b) The current rating of the secondary

Sol. (a) By transformer equation we have

$$N_s = \frac{e_s}{e_p} \times N_p$$

$$\Rightarrow \quad N_s = \left(\frac{4.4 \times 1000}{220}\right) \times 1000 = 20000 \text{ turns}$$

(b) Power supplied at primary coil is given as

$$i_p e_p = 6.6 \times 10^3 W$$

$$\Rightarrow \quad i_p = \frac{6.6 \times 10^3}{220} = 30A$$

For an ideal transformer we use

$$\frac{i_t}{i_p} = \frac{N_p}{N_s} = \left(\frac{1000}{20,000}\right) = \frac{1}{20}$$

$$\Rightarrow \quad i_s = \left(\frac{1}{20} \times i_p\right) = 1.5A$$

Q. **In a transformer there are 10000 turns in primary coil and 25000 turns in secondary coil. An AC EMF e = 50sinπt is applied across primary coil, find the peak EMF across secondary coil in ideal conditions.**

Sol. As per given turns ratio, this is a 1:2.5 step up transformer. Using transformer equation we have

$$\frac{e_1}{e_2} = \frac{N_1}{N_2}$$

$$\Rightarrow \quad e_2 = \frac{N_2}{N_1} \times e_1 = \frac{25000}{10000} \times 50V$$

$$\Rightarrow \quad e_2 = 125V$$